Hoang Hieu Long

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Last updated: November 13, 2026

Profile

Electrical engineer with research and industry experience since 2022, specialising in cable systems, design optioneering and power-system studies. At OWC – an Aqualis Company, contributed to 20+ offshore wind, solar PV, BESS and HVAC interconnector projects from feasibility to FEED, led three electrical work packages and built optimisation tools used on live projects. Previously led a five-member hydrogen and microgrid research team at HUST. Currently studying MSc Power Systems Engineering at UCL and consulting for OWC.

Power systems:	Cables (33–275 kV), substations (230–500 kV), reactive power, microgrids
Techno-economics:	Optioneering on losses, CAPEX and NPV; due diligence; route to market
Software:	Python, DIgSILENT PowerFactory, MATLAB, VBA, QGIS, AutoCAD, LaTeX
Languages:	English (IELTS 7.0), Vietnamese (native)

Education

University College London (UCL), **United Kingdom** 2026–2027
MSc Power Systems Engineering

Hanoi University of Science and Technology (HUST), **Vietnam** 2020–2025
BSc Electrical Engineering; top 10% of the cohort; thesis completed 2024, degree awarded 2025
Graduation thesis 10/10, ranked 1st of 200+ students.

Shibaura Institute of Technology (SIT), **Japan** Apr–Nov 2023
Research exchange, Electromagnetic Actuator Laboratory

Professional and Research Experience

Industry

Consultant – Electrical Engineer (zero-hours, remote), OWC – an Aqualis Company 2026–Present

- HV and LV analysis, calculations and advice on renewable energy projects in owner's engineer, technical due diligence and power-system desktop study roles.
- Develop and maintain analytical software tools for the electrical team.

Electrical Engineer, OWC – an Aqualis Company (formerly ABL Group), Vietnam 2024–2026

- **Scope:** designed and supported 20+ utility-scale projects (50 MW to 2 GW) in offshore wind, solar PV, BESS and interconnectors across the UK, Europe, MENA, APAC and the US, from feasibility to FEED.
- **Cable systems and substations:** electrical design of 33 kV PV and BESS systems, 66–132 kV inter-array systems and export cables up to 275 kV, plus concept design of 230–500 kV substations.
- **Power-system studies and techno-economics:** reactive power and losses for HVAC submarine cables over 100 km in PowerFactory; cost estimates, NPV option studies, technical due diligence and route-to-market assessments.
- **Leadership:** led 3 electrical work packages, coordinating 5+ multidisciplinary team members.
- **Software development:** built engineering tools in Python, VBA and QGIS, introduced at WindEurope and used on 5+ projects; managed and quality-checked 5 in-house design tools used on 10+ projects a year.

Research

Hydrogen Team Leader, 100% Renewable Energy Laboratory, HUST 2022–2025
Supervisor: Associate Professor Nguyen Duc Tuyen

- Led a five-member team researching microgrid energy management and hydrogen systems.
- **Energy management:** built MATLAB models of a microgrid with battery–hydrogen storage and grid-forming control, including a state-space model for MPC benchmarked against fuzzy-logic and state-machine control.
- **Hydrogen systems:** PV–electrolyser direct coupling; Power-to-Hydrogen life-cycle emissions.
- **Policy:** wrote 5 policy briefs comparing US and Vietnamese hydrogen policy with global trends; led a green city model with hydrogen storage, funded with USD 5,000 by Green In.

First Author, Book Chapter (accepted, in preparation), ASEAN Centre for Energy 2026

- Building a project risk framework and offshore wind financial model that tests how grid delay, curtailment, price and financing affect NPV, required tariff and debt-service capacity in Viet Nam and the Philippines.

Selected Projects

1.32 GW offshore wind, Poland – *lead engineer, inter-array cable package* FEED
Set turbines per string from losses and the cable schedule, and compared layouts with and without an offshore substation on NPV; client feedback of 9/10.

500–1,000 MW offshore wind, Morocco – *lead engineer, electrical package* Feasibility
Compared four turbine and layout options on losses, cable schedule and NPV, including 66 kV and 132 kV inter-array cost estimates.

Up to 1,200 MW offshore wind export, South Korea – *export optioneering* Pre-FEED
Compared HVAC with multiple offshore substations, HVAC with a single offshore substation and HVDC over about 150 km of export cable, on technical risk and cost-benefit.

400 MW Indonesia–Singapore HVAC interconnector – *power-system studies* Cable design
Modelled reactive power compensation and losses for the HVAC submarine cable (over 100 km) in DIgSILENT PowerFactory.

Inter-array cable layout optimiser – *developer, Python* OWC in-house tool
Biased genetic algorithm with matrix operations and multiprocessing: one layout in 2–5 hours instead of days, with 10–30% shorter cable than the original layouts on past projects; used for array optioneering on live projects.

2 GW floating offshore wind, the Philippines – *energy market analysis* Route to market
Assessed supply and demand, transmission constraints and the generation pipeline, then turned them into an investment-risk table, mitigation measures and alternative route-to-market options.

Two onshore wind farms, Vietnam – *lead electrical reviewer* Due diligence
Reviewed grid-connection risk and network overloads under normal and N-1 conditions, with recommendations ahead of the investment decision.

Publications

Book Chapter

1. **Hoang, L.H.**, Hoang, L.-H., Nguyen, H.-G., and Nguyen, T.D. (2026). "From Offshore Wind Pipeline to Bankable Power: Managing Grid, Curtailment and Offtake Risk in ASEAN." ASEAN Centre for Energy. (Accepted, in preparation)

Journal Articles

1. Phan Van, L., **Hoang, L.H.**, and Nguyen, T.D. (2023). "A Comprehensive Review of Direct Coupled Photovoltaic–Electrolyser System." *International Journal of Hydrogen Energy*.
2. Phan Van, L., **Hoang, L.H.**, Do, C.K., Takano, H., and Nguyen, T.D. (2023). "An Improved State Machine-Based Energy Management Strategy for Renewable Energy Microgrid with Hydrogen Storage System." *Energy Reports*, 9, 194–201.

Conference Papers

1. **Hoang, L.H.**, Phan Van, L., and Nguyen, D. (2025). "Power to Hydrogen Life Cycle Assessment: Method and a Case Study for the United States." *WindEurope Annual Event*, Copenhagen.
2. **Hoang, L.H.**, Nguyen, H.A., Takano, H., Fujita, G., and Nguyen, T.D. (2024). "A Small Scale Microgrid Experimental Model Integrated Battery–Hydrogen Storage System." *IEEJ*, Japan.
3. Do, C.K., Nguyen, H.A., **Hoang, L.H.**, Phan Van, L., Takano, H., and Nguyen, T.D. (2024). "Grid Forming-Based Hydrogen System for High-Penetration Renewable Energy Microgrid." *CPESE*, Japan.

Honours and Awards

OWC Scholarship, top 5 of 200 students in the School 2024
Third Prize, Student Forum, top 30 of 1,000+ student research papers in Vietnam 2022
Third Prize, Vietnam Mathematical Olympiad, top 45 of 2,000+ in the national round

Other Experience

Presenter, 100% Renewable Energy Laboratory research seminars 2023–2025
Delegate, Youth Engagement in Just Energy Transition, UNDP Vietnam 2023
Secretary, HUST Youth Union 2020–2024